

CS3630: Intro to Robotics and Perception

Lab 4 - Behaviour Cloning

Behavioral cloning is a simple form of imitation learning in which a policy is trained with supervised learning to mimic an expert's actions. In this lab, you will use expert demonstrations collected from pretrained policies and train a neural network policy to imitate them on continuous-control tasks from OpenAI Gym. These control tasks are simulated in MuJoCo, a physics engine commonly used for continuous-control robotics problems. One of the main environments in this assignment is the Ant task, in which a four-legged robot must learn to coordinate its joints to move forward efficiently without falling over. Because the Ant has a high-dimensional continuous action space and complex body dynamics, it is a useful benchmark for studying how well behavioral cloning can reproduce expert locomotion behavior.

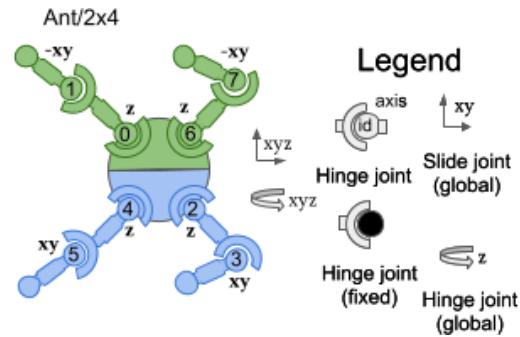
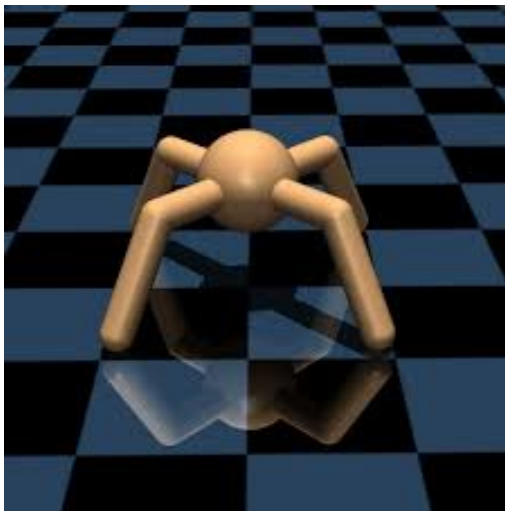


Figure 1: Two views of the MuJoCo Ant task used in this lab. The agent controls a four-legged robot and must learn stable forward locomotion.

Your work for this lab will be evaluated using the following rubric:

- **Implementation correctness:** 40 pts
- **Training checkpoint evaluation and policy behavior:** 20 pts
- **Reflection questions:** 10 pts each - total 30pts

1. Behavioral Cloning code

1. Set up the assignment environment. You may complete the lab either on Google Colab or on your own machine. See the homework README.md file for additional background and implementation details, as well as tips on how to tune the params.
2. Complete the code by filling in all sections marked with TODO. In particular, see:
 - `lab4/infrastructure/rl_trainer.py`
 - `collect_training_trajectories`
 - `train_agent`

- lab4/policies/MLP_policy.py
 - get_action
 - implement the update method in MLPPolicySL
- lab4/infrastructure/replay_buffer.py
 - sample_random_data
- lab4/infrastructure/utils.py
 - sample_trajectory
 - sample_trajectories
 - sample_n_trajectories
- lab4/infrastructure/pytorch_util.py
 - build_mlp

2. Reflection Questions

1. Describe how your learned policy behaves in one of your rollout videos. What does the agent appear to do well, where does it struggle, and how do its behaviors help explain the quantitative performance you observed?
2. In your own words, explain your behavioral cloning implementation in this lab. How does the learned policy fit the expert data, and what objective is it optimizing during training?
3. Describe one or more scenarios that you think behavioral cloning is likely to perform poorly. Why does it struggle in those settings, and what ideas or techniques could help address these limitations? (list at least one approach)

3. Turning it in

- Turn in your assignment on Gradescope. Upload the zip file with your code and log files to Lab4 Code, and answer your questions on the Gradescope questions part.
- Make sure that the submitted folder contains the implemented .py files that you implemented above and your run checkpoints under /data.
- Please go to TA Office Hours for questions.